

70 NEET-style Organic Chemistry MCQs

1. Which effect increases electron density through pi-bond delocalization? A. +I B. -I C. +M D. -M
2. Most stable carbocation is: A. CH_3^+ B. 1deg C. 2deg D. 3deg
3. Increasing acidity order: A. Phenol < Ethanol < Acetic acid < p-Nitrophenol B. Ethanol < Phenol < p-Nitrophenol < Acetic acid C. Acetic acid < Phenol < Ethanol < p-Nitrophenol D. Phenol < p-Nitrophenol < Ethanol < Acetic acid
4. Hyperconjugation is absent in: A. $(\text{CH}_3)_3\text{C}^+$ B. $\text{CH}_3\text{-CH}_2^+$ C. Benzyl^+ D. CH_3^+
5. Hybridization of carbon in benzene is: A. sp B. sp^2 C. sp^3 D. dsp²
6. Aromaticity depends on: A. Pi electrons only B. Closed ring obeying $4n+2$ rule C. Heteroatom presence D. Alternating single bonds
7. Anti-aromatic compound is: A. Cyclobutadiene B. Benzene C. Naphthalene D. Anthracene
8. Major product of alkyne hydrogenation with Lindlar catalyst is: A. Trans-alkene B. Alkane C. Cis-alkene D. Radical product
9. Free radical substitution is shown by: A. HBr addition B. Methane chlorination C. Benzene nitration D. Ethene hydration
10. Anti-Markovnikov addition occurs in: A. Hydration of alkene B. HBr/ROOR C. HBr only D. Nitration
11. Friedel-Crafts alkylation requires catalyst: A. FeCl_3 B. AlCl_3 C. Zn/Hg D. Pd-BaSO₄
12. Ozonolysis of ethene gives: A. CH_3CHO B. HCHO C. HCOOH D. $\text{CO}_2 + \text{H}_2\text{O}$
13. SN1 rate depends on: A. Nucleophile B. Leaving group & carbocation stability C. Solvent polarity only D. 1,2-shift rearrangement
14. SN2 reaction shows: A. Racemisation B. Retention C. Inversion D. No mechanism
15. Most reactive for SN2 is: A. 3deg halide B. 2deg halide C. 1deg halide D. Aryl halide
16. Vinyl halide fails SN1 because: A. No alpha-H B. Carbocation unstable C. Pi restricts back attack D. Steric crowding only
17. Benzyne intermediate appears in: A. SN1 B. SN2 C. EAS D. Aryl halide substitution
18. Alcohol has higher boiling point than alkane due to: A. Inductive effect B. LDF C. Hydrogen bonding D. Resonance
19. Iodoform test is positive for: A. Methanol B. $\text{CH}_3\text{-CO-R}$ or $\text{CH}_3\text{-CH(OH)-R}$ C. Aromatic acids D. 3deg alcohol only

20. Tertiary alcohol reacts fastest with Lucas reagent because: A. +M effect B. SN1 is favoured C. SN2 is favoured D. No rearrangement
21. Bromine water reacts with phenol giving: A. Nitro phenol B. 2,4,6-tribromophenol C. Bromobenzene D. Benzoic acid
22. Ether cleavage with conc. HI/HBr gives: A. Alcohol + alkene B. Alcohol + alkyl halide C. Diols D. Polymer
23. Cannizzaro reaction is given by aldehyde having: A. No α -H B. α -H C. 2 carbonyl groups D. γ -H
24. Fehling test is not given by: A. HCHO B. CH₃CHO C. Benzaldehyde D. CH₃-CH₂-CHO
25. Tollens test is reduced by: A. Ketones B. Aldehydes C. Alcohols D. Ethers
26. Clemmensen reduction converts: A. Carbonyl to alcohol B. Carbonyl to CH₂ (acidic) C. C-C break D. Nitro to amine
27. HVZ reaction reagent is: A. Br₂/FeBr₃ B. Br₂/P C. NBS/hv D. H₂SO₄
28. Esterification requires: A. Conc. HCl B. Mg/ether C. Alcohol/H⁺ D. NH₃
29. Most stable alkene is: A. Cis B. Trans C. gem di-substituted D. 1,2-diols
30. Grignard reagent formed with solvent: A. Water B. Ether C. Alcohol D. Acid
31. Chlorobenzene is less reactive to SN due to: A. +I effect B. C-Cl resonance C. sp³ carbon D. No pi electrons
32. Product of Wurtz reaction using CH₃Br is: A. C₂H₆ B. C₃H₈ C. C₄H₁₀ D. Benzene
33. Most basic compound is: A. NH₃ B. Aniline C. Diethyl amine D. p-nitroaniline
34. Gabriel synthesis is used to prepare: A. Aromatic amines B. Primary aliphatic amine C. Secondary amine D. Tertiary amine
35. Nitrobenzene to aniline uses: A. NaNO₂/H⁺ B. H₂/Pd C. Sn/HCl D. ROOR
36. Cis-trans isomerism shown in: A. Ethane B. 2-Butene C. Butane D. Methane
37. Aldol condensation needs aldehyde with: A. No α -H B. α -H C. Benzene D. 2 carbonyl groups
38. Reimer-Tiemann reaction product is: A. Benzoic acid B. o-hydroxybenzaldehyde C. p-cresol D. tribromophenol
39. Most stable free radical is: A. CH₃ radical B. 1^{deg} radical C. 2^{deg} radical D. 3^{deg} radical
40. Best solvent for Grignard reaction is: A. Water B. Ether C. Ethanol D. Acid

41. IUPAC name of $(\text{CH}_3)_2\text{CH}-\text{CH}_2\text{Br}$ is: A. 1-bromobutane B. 2-bromobutane C. 1-bromo-2-methylpropane D. 3-bromobutane
42. Aniline + $\text{Br}_2(\text{aq})$ gives: A. m-bromoaniline B. 2,4,6-tribromoaniline C. Bromobenzene D. No reaction
43. Decarboxylation of acetic acid gives: A. Methanol B. Methane C. Ethane D. CO_2 only
44. Williamson ether synthesis fails for: A. 1deg alkyl halide B. 2deg/3deg alkyl halide C. Aryl halide D. Methoxide
45. Order of leaving group ability is: A. $\text{F}^- < \text{Cl}^- < \text{Br}^- < \text{I}^-$ B. $\text{I}^- < \text{Br}^- < \text{Cl}^- < \text{F}^-$ C. $\text{Cl}^- < \text{F}^- < \text{I}^- < \text{Br}^-$ D. Random order
46. Ozonolysis breaks: A. C-C sigma only B. $\text{C}=\text{C}$ pi + sigma C. C-H D. C-X
47. Major product of benzene nitration is: A. m-nitrobenzene B. o-nitrobenzene C. p-nitrobenzene D. Nitrobenzene
48. Alkane carbon is hybridized as: A. sp B. sp^2 C. sp^3 D. sp^3d
49. $\text{S}_\text{N}1$ rate is highest for: A. CH_3X B. 1deg C. 2deg D. 3deg
50. Polymerization easiest for: A. Propane B. Ethene C. Ethyne D. Benzene
51. Phenol + NaOH produces: A. No reaction B. Ether C. Alkoxide D. Phenoxide
52. $\text{E}1$ elimination fastest for: A. 3deg halide B. 1deg halide C. Aryl halide D. Vinyl halide
53. Tautomerism is shown by: A. Benzene B. Acetone C. Ethanol D. Methane
54. Product of ethanol + SOCl_2 is: A. Ethyl chloride B. Ethane C. Ethene D. Acetaldehyde
55. Reaction of 1,3-butadiene + Br_2 gives: A. 1,2-addition B. 1,4-addition C. Both D. Only polymer
56. Bromination of phenol is: A. EAS B. $\text{S}_\text{N}1$ C. $\text{S}_\text{N}2$ D. Elimination
57. Keto-enol equilibrium favours enol due to: A. Steric B. H-bond C. Conjugation stability D. No $\alpha\text{-H}$
58. Hoffmann elimination major product is: A. More substituted B. Less substituted C. Alcohol D. Ketone
59. Aryl halide fails $\text{S}_\text{N}1$ because: A. Carbocation unstable B. No pi bond C. Steric D. No leaving group
60. Strongest acid is: A. p-nitrophenol B. Ethanol C. Benzene D. Methanol
61. EAS favoured when ring is: A. Activated B. Deactivated C. sp^3 D. No catalyst

62. Schiff base formed by: A. Ketone + NH_3 B. Aldehyde + 1deg amine C. Acid + amine D. Alcohol + amine
63. Ethyne hydrogenation with excess H_2 gives: A. Ethene B. Ethane C. cis-Ethene D. PVC polymer
64. Racemisation is observed in: A. $\text{S}_\text{N}2$ B. $\text{E}2$ C. $\text{S}_\text{N}1$ D. Radical
65. Diazotization temperature is: A. 25°C B. $0-5^\circ\text{C}$ C. 60°C D. 100°C
66. o/p directing group is: A. $-\text{NO}_2$ B. $-\text{CHO}$ C. $-\text{OH}$ D. $-\text{COOH}$
67. Benzene chlorination with anhydrous AlCl_3 gives: A. HCH B. Chlorobenzene C. Benzyl chloride D. meta-chloro
68. Carbylamine test by: A. 2deg amine B. 3deg amine C. 1deg amine D. Nitro
69. Boiling point order is: A. Alcohol > Alkane > Ether > Amine B. Alcohol > Amine > Ether > Alkane C. Alkane > Ether > Amine > Alcohol D. Ether > Alcohol > Amine > Alkane
70. Acidic carbonyl to CH_2 using Zn-Hg/H^+ is: A. Wolff-Kishner B. Clemmensen C. Kolbe D. Hoffmann